



Вселенная

Прикладное машинное обучение и искусственный интеллект для инженеров

Автор: Джефф Просиз

Тип файла: pdf

Размер: 20,6 МБ

Язык: Английский

Страниц: 425

Прикладное машинное обучение и искусственный интеллект для инженеров: решение бизнес-задач, с которыми не справляются традиционные алгоритмы

Введение

Инженерное дело традиционно опиралось на четко сформулированные правила, математические модели, физические законы и детерминированные алгоритмы. Если проблему можно было выразить в виде последовательности логических инструкций, инженеры обычно создавали программное обеспечение, способное ее решить.

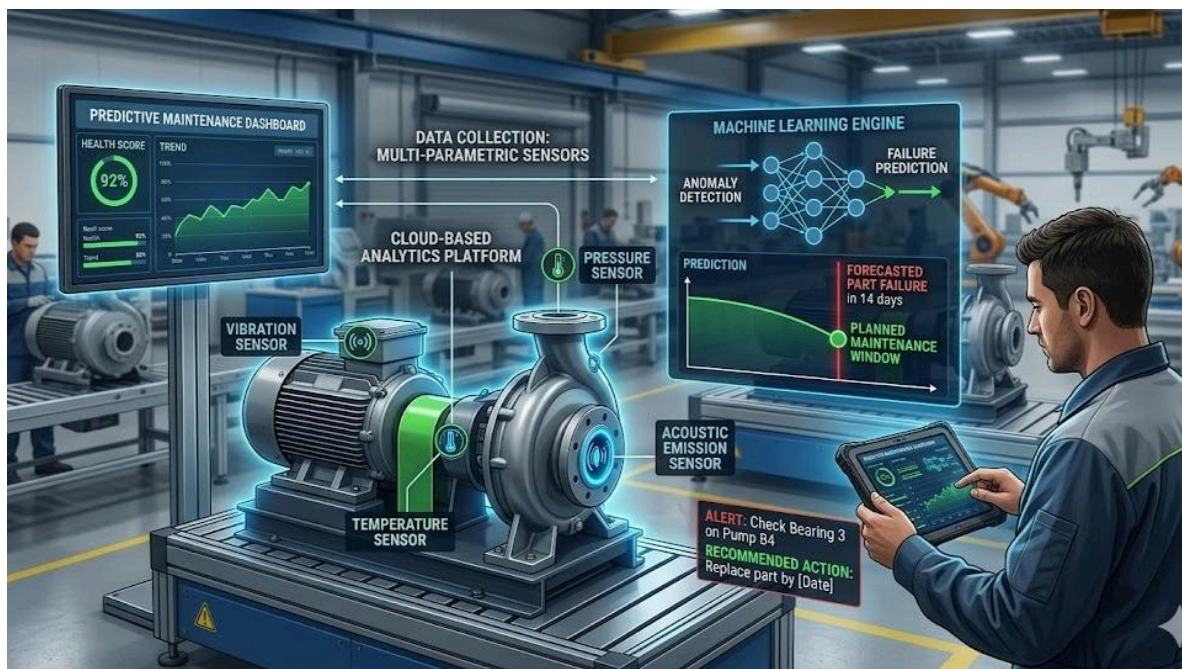
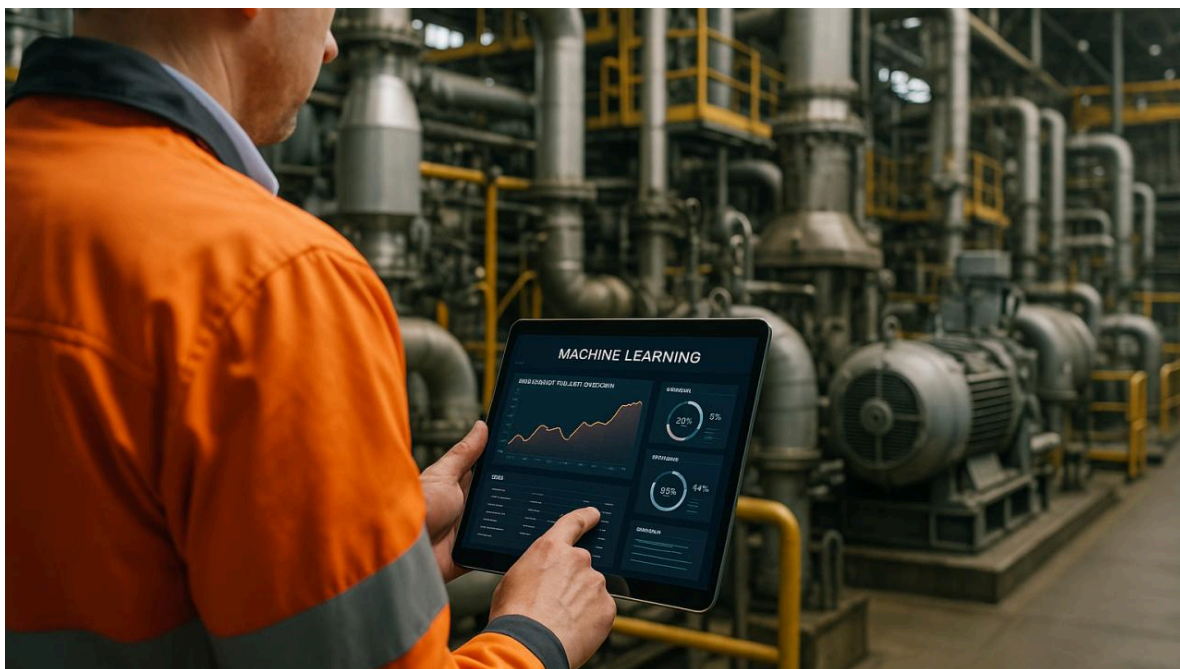
Однако современные компании все чаще сталкиваются с проблемами другого рода.

Рассмотрите такие вопросы, как:

- 🤖 Какая машина может выйти из строя на следующей неделе?
- Какой клиент с наибольшей вероятностью отменит заказ?
- Какой дефект продукта проявится в процессе производства?
- Как оптимизировать тысячи маршрутов доставки, когда условия постоянно меняются?
- На каком медицинском снимке есть отклонение от нормы?
- Какие характеристики повышают вероятность того, что клиент купит определенный товар?
- Как промышленная система может распознать необычное поведение до того, как произойдет серьезная поломка?

Эти проблемы часто **слишком динамичны, неопределенны, сложны или зависят от данных**, чтобы их можно было решить с помощью одних только традиционных алгоритмов.

Именно здесь прикладное машинное обучение (Machine Learning, ML) и искусственный интеллект (Artificial Intelligence, AI) становятся мощными инженерными инструментами. 🤖⚙️



Машинное обучение не просто заменяет традиционные инженерные методы. Напротив, оно расширяет их возможности, позволяя системам выявлять закономерности на основе исторических и непрерывно генерируемых данных.

Для студентов это открывает важную возможность: инженерное дело — это не только проектирование физических систем или написание детерминированных алгоритмов.

Оно все больше включает в себя понимание **данных, моделей, неопределенности, оптимизации, автоматизации и интеллектуального принятия решений**.

Для профессионалов главный вопрос звучит еще более практично:

Когда команде разработчиков следует использовать искусственный интеллект вместо обычного алгоритма?

Ответ зависит от структуры проблемы, имеющихся данных, цены ошибки и от того, можно ли вообще сформулировать полезные правила в явном виде.

Теоретическая основа

Традиционные алгоритмы работают исключительно хорошо, когда взаимосвязь между входными и выходными данными можно четко описать.

Например, инженерная программа может рассчитать перепад давления в известной сети трубопроводов, используя установленные формулы. Система планирования может распределять задачи в соответствии с четко заданными правилами. Калькулятор может выполнять преобразования детерминированным образом.

Машинное обучение подходит к решению этой проблемы иначе.

Вместо того чтобы явно программировать каждую взаимосвязь, инженеры предоставляют модели исторические примеры. Модель пытается выявить полезные закономерности в этих примерах, а затем применяет полученные знания к новым ситуациям.

Traditional Algorithmic Engineering

A conventional system generally follows this structure:

Input → Explicit Rules → Processing → Output

The engineer determines the rules.

Machine Learning Engineering

A machine-learning system commonly follows:

Historical Data → Learning Process → Trained Model → Prediction/Decision

The engineer designs the learning system, selects meaningful data, evaluates performance, and determines how the model should be deployed.

Why Some Problems Are Difficult to Program

Some business problems contain relationships that are:

- Highly nonlinear
- Difficult to describe mathematically
- Dependent on thousands of variables
- Continuously changing
- Influenced by human behavior
- Affected by noisy sensor data
- Dependent on visual or language information
- Too complicated for manually written rules

Machine learning is particularly useful in these situations.

Definition

Applied machine learning and AI for engineering is the practical use of data-driven learning systems to predict outcomes, classify situations, detect patterns, optimize processes, generate information, or support decisions within engineering and business environments.

The word **applied** is important.

The objective is not simply to create the most sophisticated neural network.

The objective is to **solve a meaningful problem**.

An engineer may use:

- Regression for forecasting
- Classification for decision-making
- Clustering for discovering groups
- Anomaly detection for identifying unusual behavior
- Computer vision for inspection
- Natural language processing for document analysis
- Reinforcement learning for sequential decisions
- Generative AI for engineering assistance

- Predictive models for maintenance
- Optimization combined with ML for resource allocation

AI Versus Machine Learning

AI is the broader concept of machines performing tasks associated with intelligent behavior.

Machine learning is one major approach to building AI systems.

Therefore:

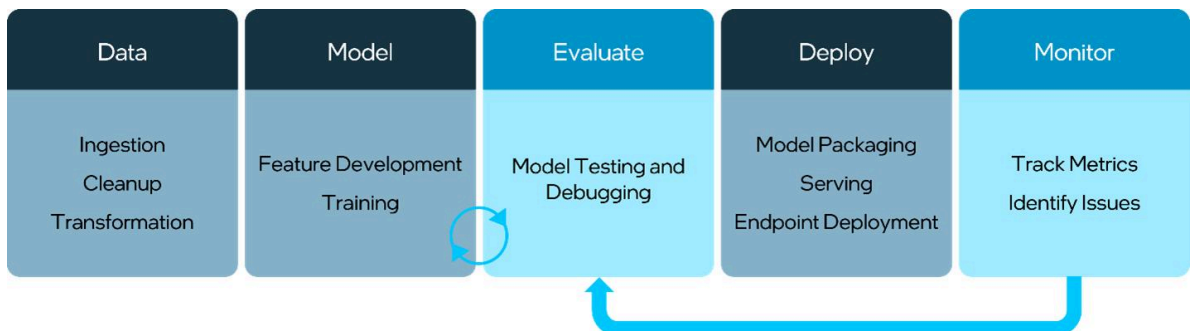
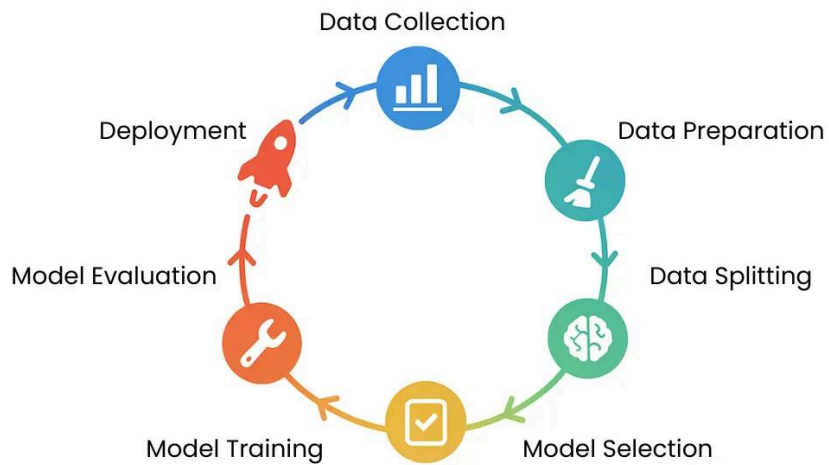
AI → Machine Learning → Algorithms + Data + Models

But modern AI also includes other techniques, including knowledge-based systems, generative models, reasoning systems, and hybrid approaches.

Step-by-Step: Turning a Business Problem Into an AI Solution



A Journey from Data Collection to Deployment



AI Model Deployment and Monitoring Process



Initiate CI/CD Pipeline

Start the automated deployment process

Ensure models meet performance criteria

Train and Validate Models



Push to Production

Deploy approved models to live environments

Revert to previous version if issues arise

Rollback on Failure

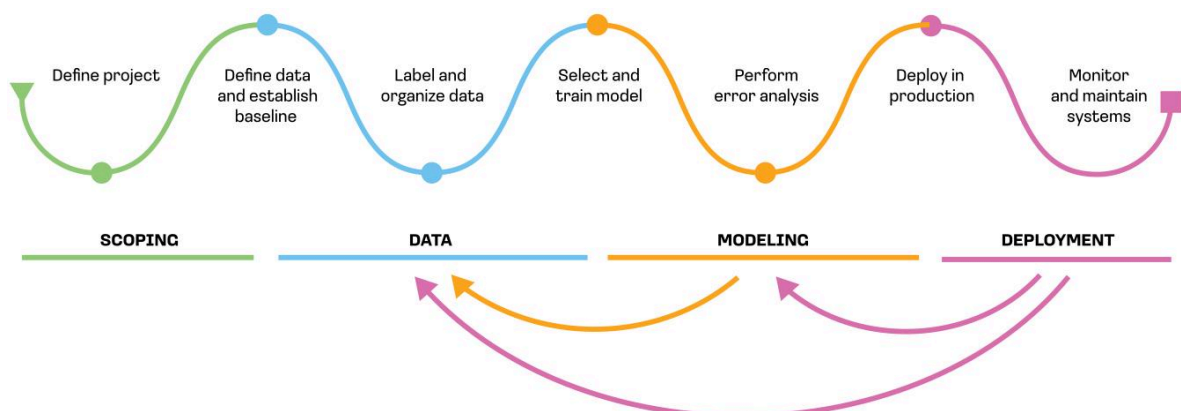


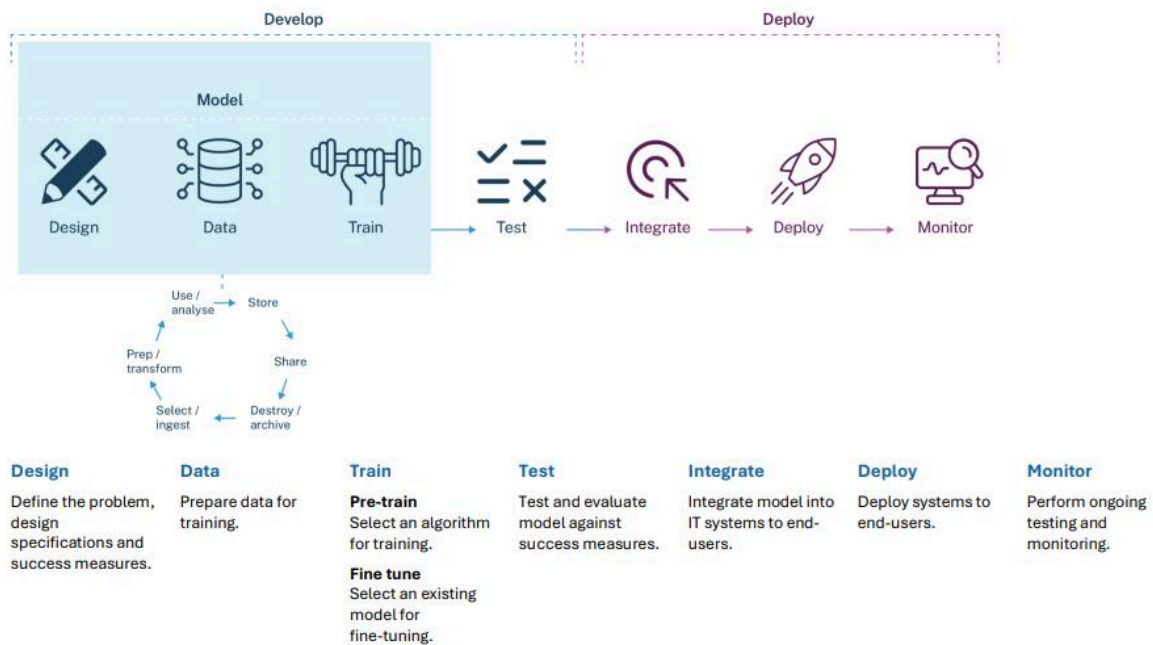
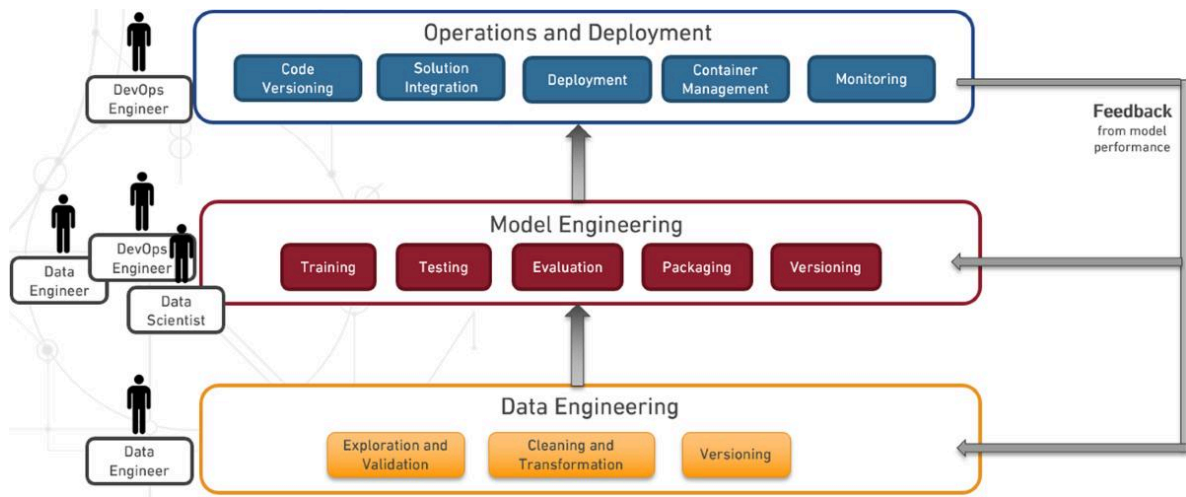
Monitor Model Drift

Use tools to detect changes in data and model performance

Deploy updates to a small user group for testing

Implement Canary Releases





Step 1: Define the Business Problem

Do not begin with:

"We need AI."

Begin with:

"What problem costs the organization time, money, safety, quality, or productivity?"

For example, a factory may experience unexpected equipment failures.

The actual business problem is not “build a neural network.”

It is:

Reduce unexpected downtime.

Step 2: Determine Whether AI Is Appropriate

Ask whether the problem can easily be solved with explicit rules.

If a simple rule produces reliable results, machine learning may be unnecessary.

For example:

If temperature exceeds a fixed safety threshold, trigger an alarm.

There is little reason to use sophisticated ML for such a deterministic condition.

However, if failure depends on temperature, vibration, operating hours, load, pressure, maintenance history, environmental conditions, and combinations of these variables, ML may become much more valuable.

Step 3: Collect Data

Data can come from:

- Sensors
- Enterprise databases
- ERP systems
- CRM systems
- Production equipment
- Cameras
- Maintenance records
- Customer interactions
- Engineering simulations
- Documents
- Logs
- IoT devices

Good AI begins with useful data.

Step 4: Prepare the Data

Real-world data is rarely perfect.

Engineers may need to handle:

- Missing values
- Duplicate records
- Incorrect measurements
- Outliers
- Different units
- Timestamp problems
- Sensor failures
- Inconsistent labels

Data preparation frequently requires more effort than model training.

Step 5: Select the Model

Model selection should reflect the problem.

A small tabular dataset may work well with tree-based methods.

Images may benefit from computer vision models.

Text may require natural language processing.

Time-series equipment data may require specialized forecasting or sequence approaches.

Step 6: Train and Validate

The model learns patterns from historical examples.

Engineers then test whether it performs effectively on data it has not previously seen.

This distinction is critical.

A model that performs brilliantly on training data but poorly on new data has **overfitted**.

Step 7: Deploy

The model can be integrated into:

- Web applications
- Manufacturing systems
- Mobile applications
- Cloud platforms
- Embedded devices
- Engineering dashboards
- Enterprise software

Step 8: Monitor Continuously

Deployment is not the final step.

Business conditions change.

Equipment changes.

Customer behavior changes.

Sensors change.

Therefore, model performance must be monitored.

This phenomenon is often called **model drift** or **data drift**.

Comparison: Traditional Algorithms vs Machine Learning

Characteristic	Traditional Algorithm	Machine Learning
Rules	Explicitly programmed	Learned from data
Best for	Clearly defined problems	Pattern-rich problems
Data requirement	Often low	Usually significant
Interpretability	Usually high	Depends on model
Adaptability	Requires rule changes	Can retrain
Complexity	Predictable	Potentially very high
Maintenance	Update logic	Monitor data and model

Characteristic	Traditional Algorithm	Machine Learning
Example	Unit conversion	Demand prediction
Failure behavior	Usually deterministic	Probabilistic

When Traditional Algorithms Win

Traditional engineering methods remain excellent when:

- Rules are known.
- Physics is well understood.
- Safety limits must be explicit.
- Exact calculations are required.
- Data is insufficient.
- The problem is simple.

When Machine Learning Wins

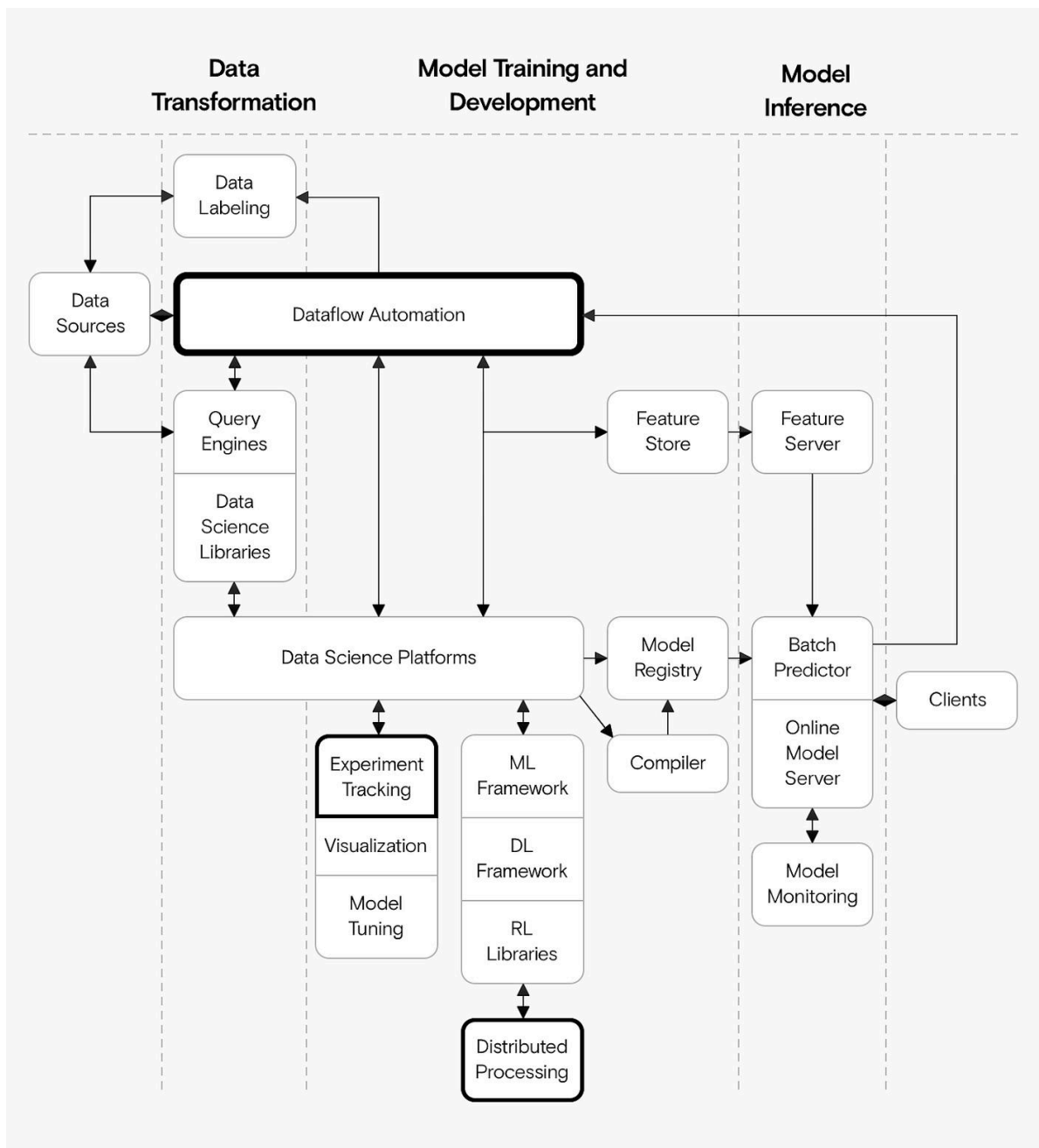
ML becomes attractive when:

- Patterns are difficult to describe manually.
- Large datasets exist.
- Relationships are nonlinear.
- The environment changes.
- Predictions are valuable.
- Human decisions are repetitive.
- Data contains hidden patterns.

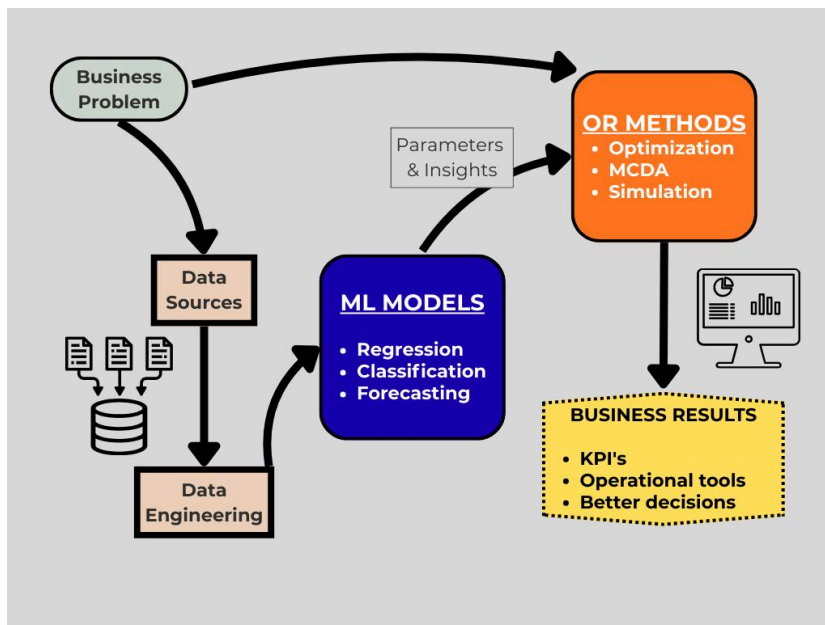
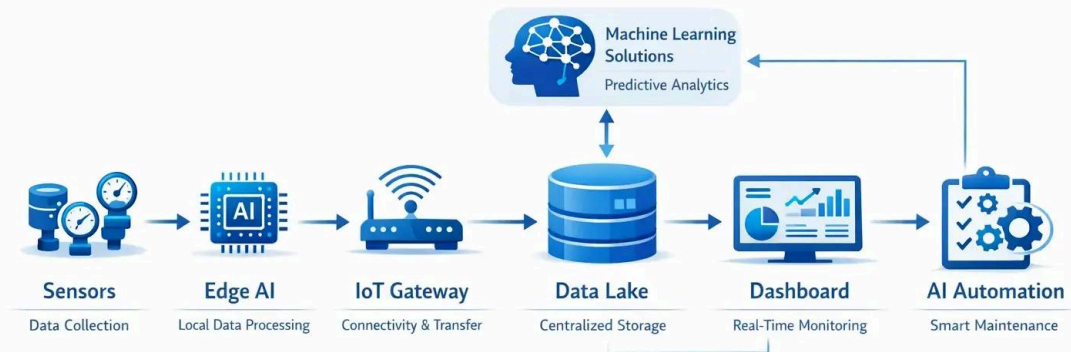
The strongest engineering systems often combine both.

Physics + Algorithms + Machine Learning = Hybrid Intelligence 🧩

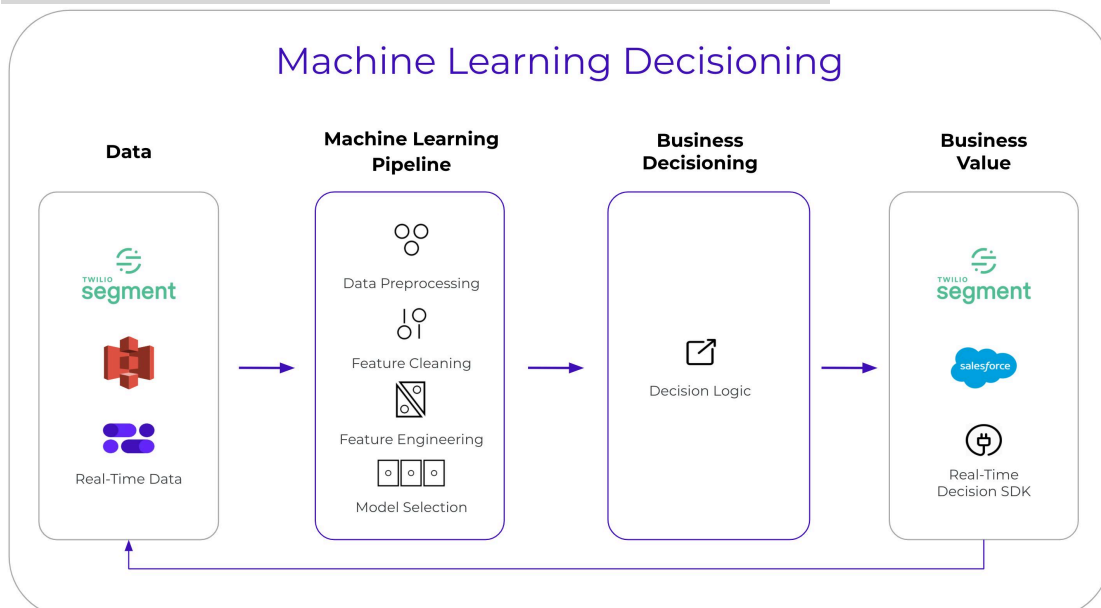
Diagrams & Engineering Frameworks

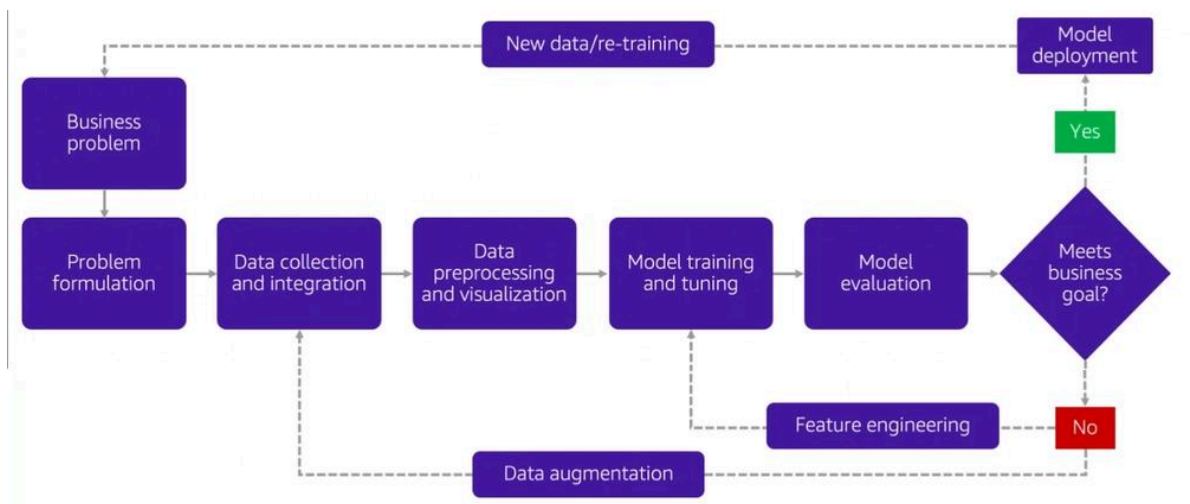
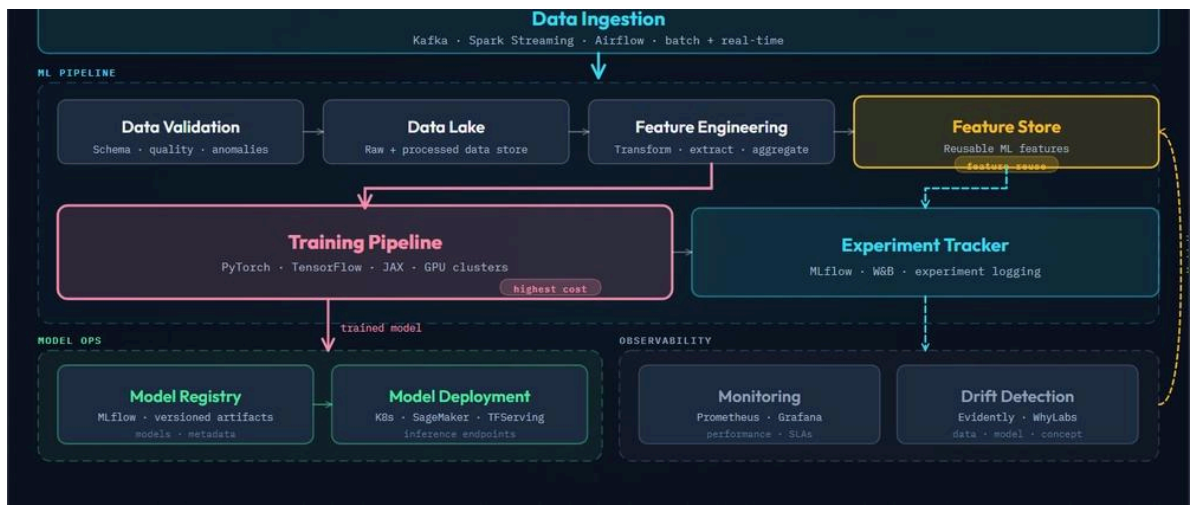


AI-Based Predictive Maintenance Architecture

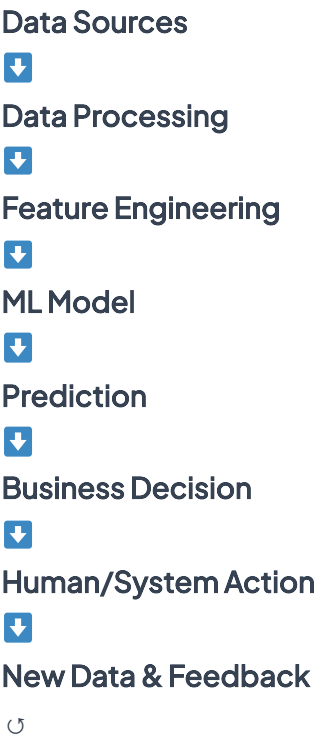


Machine Learning Decisioning





A useful engineering architecture can be visualized as:



This feedback loop is important because intelligent systems improve when engineers continuously evaluate outcomes.

AI Engineering Stack

Layer	Purpose
Data layer	Collect and store information
Processing layer	Clean and transform data
Feature layer	Represent useful signals
Model layer	Learn patterns
Evaluation layer	Measure performance
Application layer	Deliver predictions
Monitoring layer	Detect degradation
Governance layer	Control safety and compliance

Examples Without Equations

Predictive Maintenance

A manufacturing company records vibration, temperature, pressure, and operating conditions from industrial machines.

Instead of waiting for a machine to fail, an ML model identifies patterns associated with previous failures.

The maintenance team receives an early warning.

The result can be:

Less downtime + better maintenance planning + lower operational cost.

Automated Quality Inspection

A production line uses cameras to inspect components.

Traditional image processing may struggle when defects vary in shape, lighting, orientation, and appearance.

Computer vision models can learn from examples of acceptable and defective components.

Demand Forecasting

A retailer needs to estimate future demand.

Demand may depend on:

- Season
- Weather
- Promotions
- Holidays
- Location
- Customer behavior
- Competitor activity

A machine-learning model can identify interactions that are difficult to encode manually.

Customer Churn Prediction

A company wants to identify customers who may stop using its service.

Instead of applying one universal rule, ML can analyze usage patterns, support interactions, purchase history, and engagement behavior.

Real-World Applications

Applied AI is transforming engineering-heavy industries.

Manufacturing

AI can support:

- Predictive maintenance
- Defect detection
- Production optimization
- Energy management
- Inventory forecasting
- Process monitoring

Civil Engineering

Potential applications include:

- Infrastructure condition assessment
- Structural monitoring
- Construction safety
- Project delay prediction
- Material classification
- Traffic forecasting

Mechanical Engineering

Machine learning can assist with:

- Equipment diagnostics
- Failure prediction
- Design optimization
- Manufacturing control
- Energy efficiency

Electrical Engineering

Applications include:

- Load forecasting
- Fault detection
- Power quality analysis
- Grid optimization
- Renewable-energy prediction

Automotive Engineering 🚗

AI contributes to:

- Driver assistance
- Predictive maintenance
- Manufacturing inspection
- Battery management
- Traffic prediction

Energy Engineering ⚡

ML can help forecast:

- Electricity demand
- Solar generation
- Wind generation
- Equipment failures
- Energy consumption

Common Mistakes ⚠️

Starting With the Model Instead of the Problem

Choosing an advanced model before understanding the business objective often creates unnecessary complexity.

Start with the decision.

Assuming More Data Always Means Better AI

Ten million poor-quality observations may be less useful than a carefully collected dataset containing meaningful examples.

Ignoring Data Leakage

Data leakage occurs when information unavailable at prediction time accidentally enters model training.

This can create unrealistic test performance.

Measuring the Wrong Metric

High prediction accuracy does not automatically mean high business value.

For example, a rare-event detection system may achieve impressive accuracy while failing to identify the events that actually matter.

Deploying Without Monitoring

A model can degrade after deployment even if it performed well during development.

Ignoring Human Expertise

Engineers understand equipment, processes, constraints, and failure modes that may not exist in the dataset.

AI should generally complement domain expertise rather than blindly replace it.

Challenges & Solutions

Challenge	Practical Solution
Poor data quality	Build data-validation pipelines
Limited historical failures	Combine simulation, expert knowledge, and available data
Model complexity	Begin with interpretable baseline models
Changing conditions	Monitor data and retrain when appropriate
False predictions	Establish human review for critical decisions
Security risks	Apply access control and secure deployment
Lack of trust	Use explainability and transparent evaluation

Challenge	Practical Solution
High computing costs	Select efficient models and infrastructure

The Explainability Challenge

In engineering, stakeholders may ask:

“Why did the model make this prediction?”

This question is especially important in safety-sensitive industries.

Engineers should therefore consider interpretable models, feature importance, local explanations, validation tests, and carefully documented decision processes.

Case Study: AI for Predictive Factory Maintenance

Imagine an industrial manufacturer operating hundreds of machines.

Historically, maintenance follows a fixed schedule.

Every machine receives maintenance at predefined intervals.

The problem is that machines do not necessarily degrade according to the same schedule.

Some machines require maintenance earlier.

Others remain healthy for much longer.

The company begins collecting sensor information, maintenance records, operating conditions, and historical failure events.

Phase 1: Data Collection

The engineering team integrates data from machine sensors and maintenance systems.

Phase 2: Historical Analysis

Engineers identify patterns that occurred before previous failures.

Phase 3: Model Development

An ML system learns relationships between operational signals and failure events.

Phase 4: Testing

The model is evaluated against historical periods that were not used for training.

Phase 5: Deployment

When the model identifies a high-risk machine, the maintenance system creates an alert.

Phase 6: Human Decision

A maintenance engineer examines the alert alongside machine history and inspection information.

Phase 7: Feedback

The actual machine condition is recorded.

This feedback becomes additional information for future model improvement.

The key lesson is that **AI is not the maintenance strategy itself**.

AI becomes one component of a larger engineering decision system.

Essential Tips for Engineers

Start Small

Do not attempt to transform an entire organization with AI at once.

Select one measurable problem.

Build a Baseline

Before sophisticated ML, establish a simple benchmark.

If an advanced model cannot beat the baseline, investigate why.

Involve Domain Experts

A data scientist may understand models deeply, while a mechanical engineer may understand the machine deeply.

The best solutions combine both perspectives.

Focus on Business Value

Track outcomes such as:

- Reduced downtime
- Lower cost
- Improved quality
- Increased productivity
- Reduced energy consumption
- Faster decision-making

Design for Deployment

A model that exists only inside a notebook is not yet an engineering solution.

Consider:

Reliability → Latency → Security → Monitoring → Maintenance

Treat AI as an Engineering System

A production AI solution should have:

Inputs → Processing → Model → Decision → Feedback → Monitoring

This mindset makes AI much more manageable.

FAQs ?

What is [applied machine learning](#) in engineering?

Applied machine learning is the use of data-driven models to solve practical engineering and business problems such as prediction, classification, anomaly detection, optimization, and automated inspection.

Is machine learning better than traditional algorithms?

Not necessarily. Traditional algorithms are usually better when rules are clearly known. Machine learning is more useful when patterns are complex, uncertain, nonlinear, or difficult to describe manually.

Do engineers need advanced mathematics to use AI?

A strong mathematical foundation is valuable, especially for advanced ML development. However, engineers can begin applying machine learning with fundamental statistics, data analysis, programming, and model evaluation skills.

Which programming language is useful for engineering AI?

Python is particularly popular because it provides a large ecosystem for data analysis, machine learning, scientific computing, visualization, and AI development.

Can AI replace engineering judgment?

AI can automate or support specific decisions, but engineering judgment remains essential, particularly where safety, physical constraints, regulations, and unusual situations are involved.

What is the biggest challenge in industrial machine learning?

One of the biggest challenges is often not the algorithm but the data. Poor measurements, incomplete records, changing operating conditions, and insufficient failure examples can significantly affect model performance.

Should every business use AI?

No. AI should be used when it provides meaningful value. A simple deterministic algorithm may be cheaper, faster, easier to validate, and more reliable than an unnecessarily complicated AI system.

What is the future of AI engineering?

The future is likely to involve increasingly integrated systems combining machine learning, generative AI, simulation, optimization, sensor networks, physics-based models, and human expertise.

Conclusion

Applied machine learning changes the way engineers approach difficult problems.

Traditional algorithms remain fundamental. They provide precision, predictability, physical constraints, and explicit logic. But some modern business problems cannot be adequately described through manually written rules.

When data contains valuable patterns that are difficult to express algorithmically, machine learning can provide another path.

The most effective engineering strategy is not:

“AI everywhere.”

It is:

“The right engineering method for the right problem.” 

For students, this means developing a combination of programming, statistics, engineering fundamentals, data analysis, and AI knowledge.

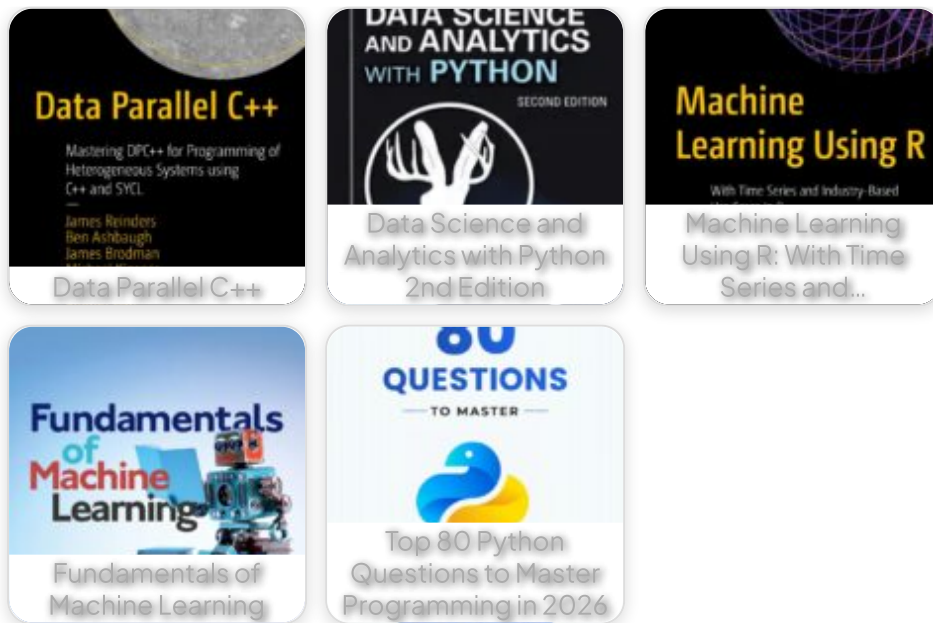
For professionals, it means learning to identify opportunities where prediction, classification, anomaly detection, optimization, computer vision, or generative AI can create measurable value.

The ultimate goal is not simply to build an intelligent model.

It is to build a **reliable engineering system that turns data into better decisions.**

That is where applied AI becomes truly powerful. 

Related Posts:



Preview PDF Book

[← Previous Book](#)

[Next Book →](#)

EngiVerse

Explore a vast library of free engineering ebooks curated to help you expand your technical knowledge, conduct groundbreaking research, and advance your career. Our mission is to create a hub of knowledge that fosters innovation, empowers aspiring engineers, and bridges the gap between education and industry.

[Home](#) [Books](#) [About Us](#) [Contact](#)
[Privacy Policy](#) [DMCA](#)